HexOct knitting: the 4H swatch

One period of the stacking A B C B A, 24 stitches by 29 rows

DICTO · knitting swatch of 31 October 2026
The idea is DICTO's: the knitted fabric is an exact edge-on slice through a stacking of balls, coloured by the pieces between the balls. Chart computed exactly with Claude (Anthropic).

What the swatch tests: whether the bias rows place the fabric where the chart assumes (each band of colour sits one third of a repeat further over, to the right going up from A to C, back to the left from C to A), and how the four colours read at this size.

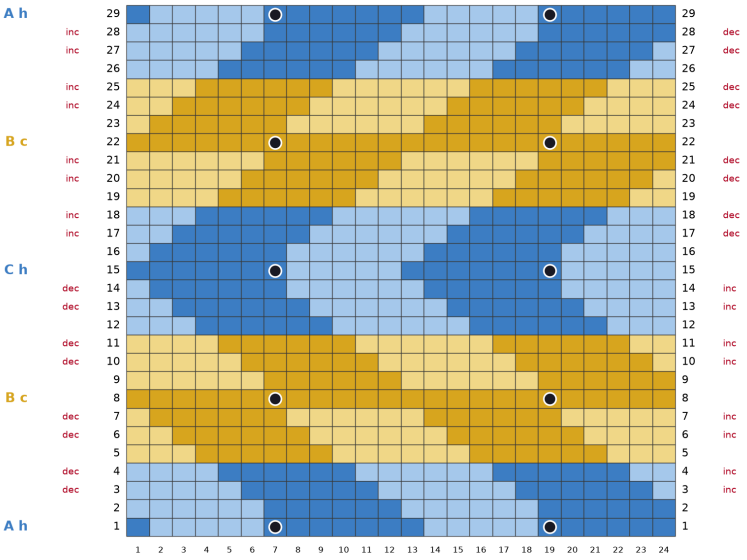
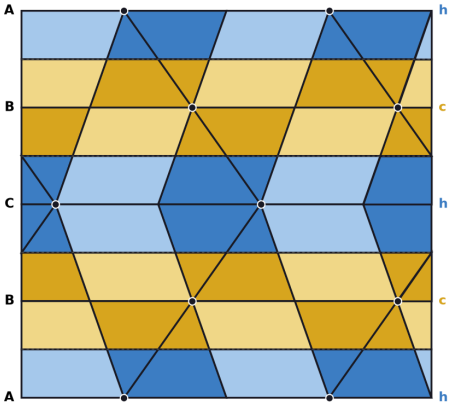


Chart. Seen from the right side, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles (ball stitches). *dec*/*inc* at the sides of a row: which edge gets the decrease and which the increase.



What you are making. The exact slice through the stacked balls (dots = ball centres, letters = layers). Gold bands belong to the c layers (B), blue to the h layers (A, C); dark = tetrahedral pieces, light = octahedral pieces. Knitted, the bias rows give the sideways steps.

Colours (4)

DG dark gold: 192 sts · LG light gold: 144 sts · DB dark blue: 186 sts · LB light blue: 174 sts

Total 696 stitches. Any yarn of one weight; four shades with clear contrast between gold and blue and between dark and light.

1. Cast on and edges

- Cast on **24 stitches** in DB (long-tail). Row 1 is the first row after the cast-on, a right-side (RS) row.
- Stockinette throughout. Stitch 1 and stitch 24 are edge stitches: knit them (RS) and purl them (WS) in the colour of the chart; do not slip them, so the edges show the bias steps.
- Colour changes: intarsia with small bobbins, twisting the yarns at every change; carry a colour along the back only where its next block is 1–2 stitches away.
- Every bias row keeps **24 stitches**: one decrease at one edge and one increase at the other, each worked one stitch in from the edge. *Bias right* (rows 3, 4, 6, 7, 10, 11, 13, 14): decrease at the left edge, increase at the right edge (as seen from the RS), so the row sits one stitch further right. *Bias left* (rows 17, 18, 20, 21, 24, 25, 27, 28): decrease at the right edge, increase at the left.

- In the colour runs below, the increased and decreased stitches are already counted as they finish (24 stitches per row); work each in the colour of its run.
- After row 29, bind off loosely in DB on the WS.

2. Bobble (MB)

In the colour of the stitch: (k1, yo, k1) into the stitch, turn, p3, turn, k3, turn, p3, turn, sl1-k2tog-psso. Rows 1, 15 and 29 are RS rows. Rows 8 and 22 are WS rows: work the bobble the same way with knit and purl exchanged ((p1, yo, p1) into the stitch, turn, k3, turn, p3, turn, k3, turn, p3tog) and push it to the RS; if it sits badly, purl these two stitches plain and sew a bobble or French knot there afterwards.

3. Row by row

DG dark gold, LG light gold, DB dark blue, LB light blue. Runs in working order (RS from stitch 24 to stitch 1, WS from stitch 1 to stitch 24); stitch numbers as on the chart.

Row	Side	Colour runs	Bias (edge stitches)	Bobble
1	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
2	WS	6 LB, 6 DB, 6 LB, 6 DB		
3	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
6	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
8	WS	24 DG		MB at st 7, 19
9	RS	6 DG, 6 LG, 6 DG, 6 LG		
10	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
11	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
12	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
13	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
14	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
15	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19
16	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB		
17	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias left**: k1, ssk, … M1, k1	
18	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: p1, M1P, … ssp, p1	
19	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
20	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
21	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
24	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
27	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
29	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19

Gauge of the design: 12 stitches per repeat, 7 rows per layer of balls, stitches about 5 wide to 4 tall (e.g. 22 stitches and 28 rows to 10 cm). Note your own gauge on the swatch; the true proportions need stitch width : row height ≈ 1.24.